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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Data Science for Engineers (course)

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Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

Week 2 ()

Week 3 ()

☐ Statistical
Modelling
(unit?
unit=47&lesso
n=48)

☐ Random
Variables and
Probability
Mass/Density
Functions
(unit?)

Week 3 : Assignment 3

The due date for submitting this assignment has passed.

Due on 2025-08-13, 23:59 IST.

As per our records you have not submitted this assignment.

1) Sumit wants to contact one of his friends, but he remembers only the first 9 of the **1 point** 10 digits of the contact number. He is sure that the last digit of the contact number is an odd number. He selects an odd number randomly. If the random variable X denotes the last digit of the contact number, then calculate $\text{Var}(X)$.

- ☐ 5
- ☐ 8
- ☐ 33
- ☐ None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
8

2) Suppose $X \sim \text{Normal}(\mu, 4)$. For $n = 20$ iid samples of X , the observed sample **1 point** mean is 5.2. What conclusion would a z-test reach if the null hypothesis assumes $\mu = 5$ (against an alternative hypothesis $\mu \neq 5$) at a significance level of $\alpha = 0.05$?

Use $F_z^{-1}(0.025) = -1.9599$

- ☐ Accept H_0
- ☐ Reject H_0

No, the answer is incorrect.
Score: 0

Accepted Answers:
 $\text{Accept } H_0$

unit=47&lesso
n=49)

☐ Sample
Statistics
(unit?
unit=47&lesso
n=50)

☐ Hypotheses
Testing (unit?
unit=47&lesso
n=51)

☐ FAQ (unit?
unit=47&lesso
n=52)

☐ Week 3
Feedback
Form : Data
Science for
Engineers!!
(unit?
unit=47&lesso
n=155)

☐ Quiz: Week 3
: Assignment
3
(assessment?
name=238)

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

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Session -
July 2025 ()

3) A box contains 8 items out of which 2 are defective. A sample of 5 items is to be selected randomly (without replacement) from the box. If the random variable X represents the number of defective items in a selection of 5 items, then find $E(X)$. (Enter the answer correct 2 decimal places) **1 point**

- ☐ 1.25
☐ 5
☐ 0.4
☐ 1.3

No, the answer is incorrect.
Score: 0

Accepted Answers:
1.25

4) Suppose $X \sim \text{Normal}(\mu, 9)$. For $n = 100$ iid samples of X , the observed sample mean is 11.8. What conclusion would a z-test reach if the null hypothesis assumes $\mu = 10.5$ (against an alternative hypothesis $\mu \neq 10.5$)? **1 point**

- ☐ Accept H_0 at a significance level of 0.10.
☐ Reject H_0 at a significance level of 0.10.
☐ Accept H_0 at a significance level of 0.05.
☐ Reject H_0 at a significance level of 0.05.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Reject H_0 at a significance level of 0.10.
Reject H_0 at a significance level of 0.05.

5) Let X and Y be two independent random variables with $\text{Var}(X) = 9$ and $\text{Var}(Y) = 3$, find $\text{Var}(4X - 2Y + 6)$. **1 point**

- ☐ 100
☐ 140
☐ 156
☐ None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
156

6) The covariance between two random variable X and Y is -3.749 , their variance is given by 3 and 5. Compute the correlation coefficient. **1 point**

- ☐ -0.854
☐ 0.561
☐ -0.968
☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

-0.968

7) When will you reject the Null hypothesis?

☐

p value greater than α

☐

p value less than α

☐

p value equal to α

☐

None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

p value less than α

8) A sample of N observations are independently drawn from a normal distribution. The sample variance follows **1 point**

☐

Normal distribution

☐

Chi-square with N degrees of freedom

☐

Chi-square with $N - 1$ degrees of freedom

☐

t-distribution with $N - 1$ degrees of freedom

No, the answer is incorrect.

Score: 0

Accepted Answers:

Chi-square with $N - 1$ degrees of freedom

9) A car manufacturer purchases car batteries from two different suppliers. Supplier X provides 55% of the batteries and supplier Y provides the rest. 5% of all batteries from supplier X are defective and 4% of all batteries from supplier Y are defective. You select a battery from the bulk and you found it to be defective. What is the probability that it is from Supplier X ? **1 point**

☐

0.0455

☐

0.455

☐

0.604

☐

0.018

No, the answer is incorrect.

Score: 0

Accepted Answers:

0.604

10) Which one of the following is best measure of central tendency for categorical data? **1 point**

☐

Mean

☐

Median

- ☐ Mode
- ☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

Mode

